

Adam Mokdad

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Education

University of Massachusetts Amherst

Bachelor of Engineering: Computer Engineering • GPA: 3.8 • Graduating Spring 2029

Relevant Coursework:

- ENGIN 112 - Intro Electrical & Computer Engineering: computer architecture, signal transmission, error detection, and PID controllers
- ENGIN 191 - Machine Learning Systems: designed, curated, and trained ML models with emphasis on performance, scalability, and system constraints
- ECE 202 - Applied MATLAB and Excel to model and solve real-world physics and electrical engineering problems, with emphasis on numerical methods, data analysis, and producing clean, well-documented code

Skills

Languages: Python, JavaScript, C, C++, C#, Lua

Machine Learning & Vision: PyTorch, TensorFlow, OpenCV, Pandas

Web & Back-end: HTML, CSS, SQL

Software Development Tools: GitHub, VS Code, MATLAB, Excel/Google Sheets, Linux/Unix CLI, Technical Documentation

Projects

NLP Database Pipeline - ENGIN 191 Final Project

- Designed and deployed an NLP pipeline using Python and PyTorch that parsed free-text input and wrote structured records to a database
- Managed the full ML pipeline: dataset curation, model training, performance evaluation, and deployment, while balancing accuracy and efficiency against system constraints

Blot (Hack Club & MIT)

- Wrote JavaScript to generate a technically complex artwork accepted into the Hack Club public Blot gallery, a collaboration with MIT's Center for Bits and Atoms
- Received an award for excellence in programming and design from Hack Club and MIT

Miscellaneous Programming Projects (GitHub)

- Independently built and published a range of projects JavaScript webtools, Unity/C# games, and Python scripts
- Advanced my skills in, computation, game development, UI/UX design, and front-end web development

Experience & Activities

Warrior Robotics Team (Nov. 2022 - Mar. 2025)

- Developed computer vision, autonomous path planning, and motor integration systems in C++ and Python
- Coordinated development across teams to deliver a competition-ready robot system
- Earned 2nd place out of 36 teams at the FIRST Robotics Reading Competition (2024)

Programming Club (Aug. 2021 - May 2024)

- Worked on collaborative and independent programming projects covering core computer science concepts
- Served as a peer tutor, supporting members with debugging, problem-solving, and code quality
- Participated in hackathons (2022 - 2023) and helped organize the annual hackathon in 2024

Computer Engineering Class (Jun. 2023 - Sep. 2024)

- Co-founded and taught an introductory programming and circuitry course for younger students in my community
- Designed hands-on lessons to make software and hardware fundamentals accessible to beginners

Volunteering: Sudbury Goodnow Public Library (Sep. 2022), Maynard OpenTable Pantry (Summer 2024), Sudbury Community Food Pantry (Summer 2024)

Awards & Honors

- AP Scholar with Distinction Award, for excellent performance on AP exams (May 2024)
- FIRST Robotics Competition, District Event Finalist (Apr. 2024)
- Excellence in programming and design with Blot, a project led by Hack Club and MIT (May 2024)
- Award for Visual Design at the Computer Science Exhibition of Lincoln-Sudbury Regional High School (May 2023)
- Award for Creativity, Innovation, and Scope of Projects in Computer Science, Lincoln-Sudbury (Jun. 2023)